PARISHRAM 2.0 2026 (Class 12th)
Biology (Botany)
Sexual Reproduction in Flowering Plants

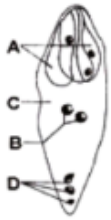
DPP: 5

Q1 In a fertilised ovule, n, 2n and 3n conditions occur respectively in:
(A) Antipodals, zygote, endosperm
(B) Megaspore mother cell, nucellus,endosperm
(C) Egg, nucellus, microspore
(D) Endosperm, micropyle, egg

Q2 Double fertilization means:
(A) One of the polar nuclei fused with the synergid and one male gamete with egg cell
(B) Fusion of one of the male gametes with polar nuclei and other with antipodal cells
(C) Fusion of one of the male gametes with secondary nucleus and the other male gamete with egg cell
(D) Fusion of one antipodal cell with egg cell and synergid cell with other antipodal cell

Q3 The fusion product of secondary nucleus and male gamete is
(A) Nucellus
(B) Primary endosperm nucleus
(C) Zygote
(D) Secondary nucleus

Q4 Identify A, B, C and D structures shown in above diagram of female gametophyte–



(A) A – SynergidsB – Polar nuclei C – Central cell D – Antipodal cells
(B) A – Antipodal cells B – Polar nuclei C – Central cell D – Synergids
(C) A – Antipodal cells B – Polar nuclei C – Megas pore mother cellD – Synergids
(D) A – Filiform Apparatus B – Polar nuclei C – Central cellD – Antipodal cells

Q5 In angiosperms, pollen tube liberate their male gametes into the
(A) central cell
(B) antipodal cells
(C) egg cell
(D) synergids

Q6 Which one of the following statements regarding post-fertilization development in flowering plants is incorrect?
(A) Ovary develops into fruit
(B) Zygote develops into embryo
(C) Central cell develops into endosperm
(D) Ovules develop into embryo sac

Q7 The main embryo is developed as a result of
(A) Pollination
(B) Triple fusion
(C) Syngamy
(D) Fusion of two polar nuclei of an embryo sac

Q8 After fertilization the outer integument of ovule changes into -
(A) Testa (B) Tegmen
(C) Fruit (D) Seed

Q9 In which part of the embryo maximum growth takes place in epigeal germination: -
(A) Radicle
(B) Plumule

(C) Epicotyl
(D) Hypocotyl

Q10 The scutellum observed in a grain of wheat or maize is comparable to which part of the seed in other monocotyledons?
(A) Cotyledon (B) Endosperm
(C) Aleurone layer (D) Plumule

Answer Key

Q1	A
Q2	C
Q3	B
Q4	A
Q5	D
Q6	D
Q7	C
Q8	A
Q9	D
Q10	A

Hints & Solutions

Note: scan the QR code to watch video solution

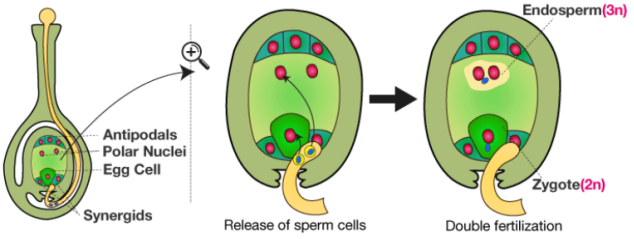
Q1 Text Solution:

Antipodals, zygote, endosperm

Video Solution:

Q2 Text Solution:

1. The process of double fertilization is unique to Angiosperms.
2. Following the release of two male gametes by the pollen tube, in the embryo sac, one male gamete near the egg fuses with the egg nucleus, resulting in the formation of a zygote.
3. The other male gamete fuses with the secondary nucleus, which results in the formation of a triploid primary endosperm mother cell.
4. Double fertilisation refers to the simultaneous occurrence of syngamy and triple fusion in the embryo sac.
5. Double fertilization is a complicated process in which one sperm cell fuses with an egg cell and the other fuses with two polar nuclei, resulting in a diploid ($2n$) zygote and a triploid ($3n$) primary endosperm nucleus (PEN).
6. Endosperm is known as triple fusion because it is the result of the fusion of three haploid nuclei.
7. The primary endosperm nucleus eventually develops into the primary endosperm cell (PEC), and then into the endosperm.



Video Solution:

Q3 Text Solution:

Primary endosperm nucleus

Video Solution:

Q4 Text Solution:

The diagram depicts a mature embryo sac with
A)Synergids at the micropylar end.
B)Polar nuclei that combine with male gametes to form endosperm.
C)Central cell with polar nuclei
D)Three antipodal cells are present at the chalazal end.
Filiform apparatus-Direct pollen tube to synergids
So Option A i.e. synergids, polar nuclei, central cell, antipodal cell is the correct answer.

Video Solution:

Q5 Text Solution:

In angiosperms, pollen tube liberate their male gametes into the synergids.

Video Solution:

Q6 Text Solution:

Ovules develop into embryo sac.

Video Solution:

Q7 Text Solution:

Syngamy is the fusion of the male and female gametes (fertilization), which leads to the formation of the zygote. The zygote develops into the main embryo.

Video Solution:

Q8 Text Solution:

After fertilization, the outer integument of the ovule develops into the testa, which is the seed coat.

Video Solution:

Q9 Text Solution:

In epigeal germination, the hypocotyl elongates and pushes the cotyledons above the ground.

Video Solution:

Q10 Text Solution:

The scutellum is the single cotyledon present in the seeds of monocotyledons like wheat or maize.

Video Solution: